

Strategic Arbitrage in the High North

n -Cell Military Organisation, Geopolitical Risk Differentials, and the Role of Indian Diplomacy in Scandinavian–Russian Tension

Soumadeep Ghosh

Kolkata, India

Abstract

This paper develops a unified theoretical framework that integrates three previously distinct bodies of knowledge: (i) the n -cell *military organisation* model, which stipulates a proportionality law between cell size and cell risk; (ii) Ghosh’s (2019) observation of an interest-rate arbitrage opportunity between the Indian and Russian economies; and (iii) the contemporary geopolitical literature on Scandinavian resilience and Nordic–Russian tensions. We formalise the n -cell proportionality constraint, prove that external shocks create measurable disequilibria in risk allocation, and show that restoring equilibrium via arms expansion is dominated—under plausible parametric conditions—by third-party diplomatic de-escalation. We then generalise Ghosh’s financial arbitrage into a *geopolitical risk arbitrage* framework, demonstrating that India’s structural position between Scandinavia and Russia constitutes an exploitable positional surplus across four channels: energy, Arctic governance, multilateral institutions, and bilateral trade. Empirical proxies, illustrative statistics, and policy algorithms are provided throughout.

The paper ends with “The End”

Contents

1	Introduction	2
2	The n-Cell Military Organisation Framework	2
2.1	Primitive definitions	2
2.2	The proportionality law	3
2.3	Shock dynamics	3
3	Scandinavia as a 3-Cell Configuration	4
3.1	Cell identification	4
3.2	Baseline proportionality check	4
3.3	Post-2022 shock analysis	4
3.4	Extension to 5-cell configuration	4
4	Geopolitical Risk Arbitrage: Generalising Ghosh (2019)	5
4.1	The original financial arbitrage	5
4.2	Generalisation to geopolitical risk differentials	5
4.3	Comparative statics	5
4.4	The Ghosh interest-rate analogy: a formal table	5
5	India’s Four Diplomatic Channels	6
5.1	Channel decomposition	6
5.2	Structural constraints	7
6	Statistical Risk Monitoring and an ML Framework	7
6.1	Disequilibrium as a signal	7
6.2	Feature engineering	7
6.3	Regression model for disequilibrium	7
6.4	A gradient-boosted early-warning algorithm	7
7	Policy Algorithm	8

8 PGF/TikZ Diagrams	9
8.1 The 3-cell Scandinavian configuration	9
8.2 Shock propagation and restoration pathways	9
8.3 India as diplomatic arbitrageur	10
8.4 Disequilibrium time series (PGFPlots)	10
9 Conclusion	10

List of Figures

1 Scandinavia as a 3-cell configuration. Cell sizes and risks are normalised; arrows denote Nordic Council coordination flows.	9
2 Restoration pathways following a geopolitical risk shock. Pathway (i) – arms expansion (dashed red) – risks an arms-race loop when $\lambda \neq 1$. Pathway (ii) – diplomatic de-escalation (solid teal) – achieves stable equilibrium under Theorem 2.7.	9
3 India as diplomatic arbitrageur between the Scandinavian bloc and Russia. India’s positional surplus Π_{geo} is positive provided both bilateral linkages carry non-trivial value (Proposition 4.3).	10
4 Stylised disequilibrium index $\hat{\delta}_i(t)$ for each Scandinavian cell, 2018–2025. The 2022 shock sharply widens all indices; Sweden’s cell experiences the largest displacement, consistent with equations (8)–(10).	10

List of Tables

1 Scandinavian 3-cell configuration: empirical proxies (2024)	4
2 Formal analogy: Ghosh (2019) financial arbitrage $\leftrightarrow$ geopolitical risk arbitrage	6
3 India’s four diplomatic channels: characteristics and leverage	6
4 Illustrative early-warning model performance (stylised)	8

1 Introduction

The post-2022 security environment in Northern Europe has produced a sharp increase in geopolitical risk across the Scandinavian peninsula. Sweden and Finland’s accession to NATO (2023–24), accelerated Nordic defence integration, and Russia’s persistent hybrid-warfare campaigns have collectively disrupted a half-century of managed low tension in the High North. At the same time, India has deepened its economic and strategic relationships with both Russia and the Nordic bloc, occupying a positional node that no Western actor can claim.

This paper argues that these phenomena are formally connected. The n -cell military organisation model—which distributes strategic mass and risk across national sub-units—provides a structural language for describing the Scandinavian bloc. Ghosh’s (2019) financial arbitrage insight—that interest-rate differentials between Russia and India create exploitable spreads—generalises naturally into a *geopolitical risk differential* that India is uniquely positioned to intermediate. The combination yields a novel policy prescription: Indian diplomatic engagement reduces the risk load on Scandinavian cells more efficiently than endogenous military expansion, and does so while preserving India’s own strategic and commercial interests.

The paper is structured as follows. Section 2 formalises the n -cell framework and proves key properties. Section 3 applies the framework to Scandinavia as a 3-cell configuration. Section 4 develops the geopolitical arbitrage extension of Ghosh (2019). Section 5 analyses India’s four diplomatic channels. Section 6 presents a stylised machine-learning approach to risk monitoring. Section 7 derives the policy algorithm. Section 9 concludes.

2 The n -Cell Military Organisation Framework

2.1 Primitive definitions

Definition 2.1 (n -cell military). A state’s military $\mathcal{M}$ of total size $M \in \mathbb{R}_{>0}$ is partitioned into n non-overlapping *cells* $\{c_1, c_2, \dots, c_n\}$, where $c_i \in \mathbb{R}_{>0}$ denotes the strategic mass (personnel, materiel, and economic capacity) of the i -th cell, such that

$$M = \sum_{i=1}^n c_i. \quad (1)$$

Definition 2.2 (Cell risk). Let $R_i \in \mathbb{R}_{>0}$ denote the *operational risk* borne by cell i , and let $R \in \mathbb{R}_{>0}$ denote total systemic risk. Then

$$R = \sum_{i=1}^n R_i. \quad (2)$$

2.2 The proportionality law

The fundamental law of n -cell organisation asserts that an efficient military configuration satisfies a proportionality constraint across all cells.

Definition 2.3 (Proportional configuration). The military is in a *proportional configuration* if and only if

$$\frac{c_i}{R_i} = \frac{M}{R} \quad \forall i \in \{1, \dots, n\}. \quad (3)$$

Equation (3) states that each cell's *risk-adjusted mass ratio* equals the bloc's global ratio. We now establish key mathematical properties of this configuration.

Proposition 2.4 (Equivalence to proportional risk allocation). *Equation (3) holds if and only if $R_i = \frac{c_i}{M} \cdot R$ for all i .*

Proof. ($\Rightarrow$) Suppose (3) holds. Then $c_i/R_i = M/R$, so $R_i = c_i R/M = (c_i/M) \cdot R$.

($\Leftarrow$) Suppose $R_i = (c_i/M) \cdot R$. Then $c_i/R_i = c_i/((c_i/M) \cdot R) = M/R$. Consistency with (2) follows since $\sum_i R_i = (R/M) \sum_i c_i = (R/M) \cdot M = R$. $\square$

Definition 2.5 (Disequilibrium index). For cell i , define the *disequilibrium index*

$$\delta_i = \frac{c_i}{R_i} - \frac{M}{R}. \quad (4)$$

A proportional configuration satisfies $\delta_i = 0$ for all i . $\delta_i < 0$ implies cell i is *over-exposed* (risk exceeds its proportional share); $\delta_i > 0$ implies *under-exposure*.

Proposition 2.6 (Zero-sum disequilibrium). *The weighted sum of disequilibria vanishes: $\sum_{i=1}^n R_i \delta_i = 0$.*

Proof. $\sum_{i=1}^n R_i \delta_i = \sum_{i=1}^n R_i \left(\frac{c_i}{R_i} - \frac{M}{R} \right) = \sum_{i=1}^n c_i - \frac{M}{R} \sum_{i=1}^n R_i = M - \frac{M}{R} \cdot R = 0$. $\square$

2.3 Shock dynamics

Let $\varepsilon_i(t) \geq 0$ denote an exogenous risk shock to cell i at time t . The post-shock risk is $R_i(t) = R_i^* + \varepsilon_i(t)$, where R_i^* is the equilibrium risk.

Theorem 2.7 (Restoration pathways). *Following a positive shock $\varepsilon_i(t) > 0$ to cell i , the proportionality law (3) can be restored by exactly two classes of intervention:*

- (i) **Internal (arms expansion):** increase c_i to $c'_i = c_i + \Delta c_i$ such that $\Delta c_i = c_i \varepsilon_i(t)/R_i^*$.
- (ii) **External (diplomatic de-escalation):** reduce $\varepsilon_i(t)$ to zero through third-party intervention, leaving c_i unchanged.

Proof. Under pathway (i): the new ratio $c'_i/R_i(t) = (c_i + \Delta c_i)/(R_i^* + \varepsilon_i) = (c_i + c_i \varepsilon_i/R_i^*)/(R_i^* + \varepsilon_i) = c_i(R_i^* + \varepsilon_i)/(R_i^*(R_i^* + \varepsilon_i)) = c_i/R_i^* = M/R$. Under pathway (ii): $\varepsilon_i(t) \rightarrow 0$ directly restores $R_i(t) \rightarrow R_i^*$, hence $c_i/R_i^* = M/R$. $\square$

Remark 2.8. Pathway (i) raises M (since c_i grows), which may trigger compensating shocks ε_j in other cells via adversary reaction functions, initiating an arms-race loop. Pathway (ii) avoids this second-order effect.

3 Scandinavia as a 3-Cell Configuration

3.1 Cell identification

We model the Scandinavian strategic bloc as an $n = 3$ configuration:

$$\mathcal{M}_{\text{Scand}} = \{c_1^{\text{NO}}, c_2^{\text{SE}}, c_3^{\text{DK}}\}, \quad (5)$$

where Norway (NO), Sweden (SE), and Denmark (DK) constitute operationally distinct cells with complementary strategic domains. Table 1 provides empirical proxies for cell mass and baseline risk.

Table 1: Scandinavian 3-cell configuration: empirical proxies (2024)

Cell	Nation	Pop. (m)	GDP (\$bn)	c_i (index)	Primary risk domain
c_1	Norway	5.5	500	0.28	Arctic / energy security
c_2	Sweden	10.5	600	0.37	Baltic / land frontier
c_3	Denmark	6.0	400	0.25	North Sea / Greenland
Bloc total M			1,500	1.00	

Note: c_i index is normalised GDP weight. Risk domains sourced from Arctic Institute (2025–26) and CSIS (2025).

3.2 Baseline proportionality check

At equilibrium, Proposition 2.4 requires $R_i^* = (c_i/M) \cdot R$. Defining $R^* = 1$ (normalised), the baseline risk allocation is:

$$R_1^* = 0.28, \quad R_2^* = 0.37, \quad R_3^* = 0.25, \quad R^* = 0.90 < 1, \quad (6)$$

reflecting Finland’s $c_4 = 0.10$ residual (discussed in Section 3.4). The disequilibrium indices $\delta_i^* = 0$ by construction at baseline.

3.3 Post-2022 shock analysis

Following Russia’s February 2022 invasion of Ukraine, we model the shock vector as:

$$\varepsilon(t_0) = (\varepsilon_1, \varepsilon_2, \varepsilon_3)^\top = (0.04, 0.12, 0.06)^\top, \quad (7)$$

where the shock to c_2 (Sweden) is largest, reflecting its exposed Baltic coastline, NATO accession pressure, and direct identification as a Russian information-warfare target. The post-shock disequilibrium indices become:

$$\delta_1 = \frac{0.28}{0.32} - 1 = -0.125, \quad (8)$$

$$\delta_2 = \frac{0.37}{0.49} - 1 = -0.245, \quad (9)$$

$$\delta_3 = \frac{0.25}{0.31} - 1 = -0.194. \quad (10)$$

All cells are over-exposed ($\delta_i < 0$), confirming systemic disequilibrium. By Proposition 2.6, the weighted sum $\sum_i R_i \delta_i = 0$ remains exactly satisfied—the disequilibrium is internal to the bloc, consistent with no change in M/R at the systemic level.

3.4 Extension to 5-cell configuration

Following the paper’s own taxonomy (3-cell: Army, Navy, Airforce; 4-cell: adds Ballistics; 5-cell: adds Spaceforce), the Nordic security architecture has evolved toward $n = 5$:

$$\mathcal{M}_{\text{Nordic}} = \{c_1^{\text{NO}}, c_2^{\text{SE}}, c_3^{\text{DK}}, c_4^{\text{FI}}, c_5^{\text{Baltics}}\}, \quad (11)$$

where Finland (NATO accession 2023) and the Baltic states constitute emerging frontier cells with high R_i and relatively small c_i —cells for which the proportionality constraint is most severely violated.

4 Geopolitical Risk Arbitrage: Generalising Ghosh (2019)

4.1 The original financial arbitrage

Ghosh (2019) identifies a covered interest arbitrage opportunity between India and Russia. With $r_{RU} = 6.50\%$, $r_{IN} = 5.40\%$, and spot rate $S_0 = 1.11548$ INR/RUB, the uncovered interest parity condition is violated:

$$r_{RU} - r_{IN} = 1.10\% > 0. \quad (12)$$

A stylised covered arbitrage strategy yields gross return (ignoring transaction costs):

$$\Pi_{\text{fin}} = S_0 \cdot (1 + r_{RU}) \cdot \frac{1}{F_0} - (1 + r_{IN}), \quad (13)$$

where F_0 is the one-year forward rate. Ghosh’s implicit assumption is that $F_0 \approx S_0$ —true only under specific capital-control environments—making $\Pi_{\text{fin}} > 0$.

4.2 Generalisation to geopolitical risk differentials

Definition 4.1 (Geopolitical risk spread). Let $\rho_{\text{Scand}}(t)$ denote the Scandinavian bloc’s aggregate geopolitical risk premium and $\rho_{RU}(t)$ Russia’s international-isolation risk premium. The *geopolitical spread* is:

$$\Sigma(t) = \rho_{\text{Scand}}(t) - \rho_{RU}(t). \quad (14)$$

Definition 4.2 (Diplomatic arbitrage position). A third-party state $\mathcal{P}$ holds a *diplomatic arbitrage position* if it maintains simultaneously:

- (i) credible economic linkages to Russia (“long Russia”), and
- (ii) credible strategic partnerships with the Scandinavian bloc (“long Scandinavia”),

such that $\mathcal{P}$ ’s expected payoff from tension reduction exceeds its expected payoff from either pure alignment.

Proposition 4.3 (India’s arbitrage surplus). Let V_{IN-RU} be the value of India’s Russia linkage and V_{IN-SC} the value of India’s Scandinavian linkage. Define the diplomatic arbitrage surplus:

$$\Pi_{\text{geo}} = V_{IN-RU} + V_{IN-SC} - \max(V_{IN-RU}, V_{IN-SC}). \quad (15)$$

Then $\Pi_{\text{geo}} = \min(V_{IN-RU}, V_{IN-SC}) > 0$ provided both linkages are non-trivially valuable.

Proof. $\Pi_{\text{geo}} = V_{IN-RU} + V_{IN-SC} - \max(V, W)$ where $V = V_{IN-RU}$, $W = V_{IN-SC}$. Without loss of generality suppose $V \geq W > 0$. Then $\Pi_{\text{geo}} = V + W - V = W = \min(V, W) > 0$. $\square$

Remark 4.4. Proposition 4.3 formalises the intuition that India’s *positional surplus* derives from simultaneous engagement with both parties—a structure directly analogous to the bid–ask spread captured by a financial intermediary.

4.3 Comparative statics

Differentiating (14) with respect to the shock vector ε :

$$\frac{\partial \Sigma}{\partial \varepsilon_i} = \frac{\partial \rho_{\text{Scand}}}{\partial \varepsilon_i} > 0, \quad (16)$$

since higher Scandinavian risk widens the spread, *increasing India’s diplomatic surplus*. This creates a structural incentive for India to prefer moderate—but non-zero—levels of tension, consistent with observed Indian diplomacy.

4.4 The Ghosh interest-rate analogy: a formal table

Table 2 maps every variable of Ghosh’s 2019 framework onto its geopolitical counterpart.

Table 2: Formal analogy: Ghosh (2019) financial arbitrage $\leftrightarrow$ geopolitical risk arbitrage

Concept	Financial (Ghosh 2019)	Geopolitical (this paper)
Rate A	Russia bank rate 6.50%	Russian isolation risk premium
Rate B	India bank rate 5.40%	Nordic geopolitical risk premium
Spread	$r_{RU} - r_{IN} = 1.10\%$	$\Sigma = \rho_{Scand} - \rho_{RU}$
Spot rate	1 RUB = 1.11548 INR	Diplomatic exchange rate
Arbitrageur	Indian citizen / firm	Republic of India (state actor)
Instrument	Currency + deposits	Diplomatic channels (4 types)
Return II	Interest differential	$\Pi_{geo} = \min(V_{IN-RU}, V_{IN-SC})$
Transaction cost	Capital controls, FX fees	Credibility gap, abstentions

5 India’s Four Diplomatic Channels

5.1 Channel decomposition

We decompose India’s total diplomatic value into four additive channels:

$$V_{IN-RU} + V_{IN-SC} = \sum_{k=1}^4 V_k, \quad (17)$$

where V_1 (energy), V_2 (Arctic), V_3 (institutional), and V_4 (trade/technology) are defined below and summarised in Table 3.

Channel 1 – Energy. India’s absorption of Russian crude at a discount (≈ 1.7 – 2.0 million barrels per day, 2023–24) sustains Russian export revenue while signalling to Moscow that India is an indispensable partner. The indirect Scandinavian benefit operates through Russia’s reduced incentive to weaponise energy routes in the North Sea and Baltic.

Channel 2 – Arctic governance. India holds Observer status at the Arctic Council. The Council’s gradual resumption of operations (Russia re-admitted to working groups in early 2024) provides India with a multilateral channel in which Nordic states and Russia nominally co-operate. The proposed Chennai–Vladivostok maritime corridor gives India a concrete commercial stake in Arctic stability.

Channel 3 – Institutional. India’s presence in BRICS and the Shanghai Cooperation Organisation prevents these bodies from being directed exclusively against Western interests. India’s concurrent pursuit of UN Security Council reform (G4 joint statement, September 2024) aligns with Nordic support for multilateral governance—creating a shared institutional interest that transcends the bilateral Russia–Scandinavia divide.

Channel 4 – Trade and technology. The India–Nordic bilateral encompasses defence exports, space technology, green energy transition, and maritime cooperation. As of 2024, Sweden chaired both Nordic foreign-security collaboration (N5) and the Nordic Council of Ministers, making it the natural liaison partner for formalising India’s Nordic engagement.

Table 3: India’s four diplomatic channels: characteristics and leverage

Channel	Mechanism	Russia leverage	Scandinavia leverage
V_1 : Energy	Discount oil purchases sustain Russian export revenue	High: irreplaceable buyer	Moderate: reduces energy weaponisation incentive
V_2 : Arctic	Observer status; Chennai–Vladivostok corridor	Moderate: shared governance interest	High: prevents Sino-Russian Arctic monopoly
V_3 : Institutional	BRICS/SCO moderation; UN SC reform alignment	High: India keeps bodies non-anti-Western	High: shared multilateral reform agenda
V_4 : Trade/tech	Defence, space, green energy partnerships	Low: limited direct effect	High: raises Nordic stake in India’s Russia engagement

5.2 Structural constraints

India’s arbitrage position is subject to three binding constraints that function as transaction costs in the Ghosh analogy:

- (a) **Credibility constraint:** India’s discounted Russian oil purchases and UN abstentions on Ukraine create a *credibility gap* with Nordic partners. Formally, let $\kappa \in [0, 1]$ be a credibility discount; the effective Nordic linkage value is $\kappa \cdot V_{\text{IN-SC}}$.
- (b) **Ukraine ambiguity constraint:** India’s refusal to condemn the invasion explicitly limits its ability to claim neutral broker status. This reduces κ below unity.
- (c) **China–Russia axis constraint:** Sino-Russian Arctic alignment compresses India’s manoeuvre space. If China occupies the “long Russia” position more aggressively, it reduces the exclusivity of India’s access and hence the arbitrage surplus Π_{geo} .

The constrained surplus is:

$$\tilde{\Pi}_{\text{geo}} = \min(\kappa V_{\text{IN-SC}}, \mu V_{\text{IN-RU}}), \quad (18)$$

where $\mu \in (0, 1]$ reflects the erosion of Russian linkage value as China becomes a closer substitute. India’s optimal strategy is to maximise $\tilde{\Pi}_{\text{geo}}$ subject to $\kappa, \mu > 0$.

6 Statistical Risk Monitoring and an ML Framework

6.1 Disequilibrium as a signal

The disequilibrium index $\delta_i(t)$ defined in (4) constitutes a time series whose departures from zero signal instability. We propose monitoring a rolling estimate:

$$\hat{\delta}_i(t) = \frac{\hat{c}_i(t)}{\hat{R}_i(t)} - \frac{\hat{M}(t)}{\hat{R}(t)}, \quad (19)$$

where $\hat{c}_i, \hat{R}_i$ are empirical proxies (defence spending, geopolitical risk indices such as the GPR index of Calderáa and Iacoviello).

6.2 Feature engineering

Let $\mathbf{x}_i(t) \in \mathbb{R}^p$ be a feature vector for cell i at time t , comprising:

$$\mathbf{x}_i(t) = (\text{GDP}_{i,t}, \text{DefSpend}_{i,t}, \text{GPR}_{i,t}, \text{TradeRU}_{i,t}, \text{NATOex}_{i,t}, \Delta\text{Hybrid}_{i,t})^\top, \quad (20)$$

where $\Delta\text{Hybrid}_{i,t}$ measures the increment in attributed hybrid-warfare incidents.

6.3 Regression model for disequilibrium

We fit a linear regression:

$$\hat{\delta}_i(t) = \beta^\top \mathbf{x}_i(t) + \eta_i + \epsilon_{i,t}, \quad \epsilon_{i,t} \sim \mathcal{N}(0, \sigma^2), \quad (21)$$

with cell fixed effects η_i and i.i.d. errors. The OLS estimator $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ is unbiased under standard assumptions.

6.4 A gradient-boosted early-warning algorithm

For early warning of escalation, we define a binary outcome $y_{i,t} \in \{0, 1\}$ indicating whether $|\delta_i(t)| > \tau$ for threshold $\tau > 0$. A gradient-boosted classifier iterates:

$$F_0(\mathbf{x}) = \arg \min_{\gamma} \sum_{i,t} L(y_{i,t}, \gamma), \quad (22)$$

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu h_m(\mathbf{x}), \quad m = 1, \dots, T, \quad (23)$$

where L is cross-entropy loss, $\nu \in (0, 1]$ is the learning rate, and h_m is a weak learner (shallow decision tree) fit to the negative gradient (pseudo-residuals):

$$r_{i,t}^{(m)} = - \left[\frac{\partial L(y_{i,t}, F(\mathbf{x}_{i,t}))}{\partial F(\mathbf{x}_{i,t})} \right]_{F=F_{m-1}}. \quad (24)$$

The feature importance scores from this model identify which components of $\mathbf{x}_i(t)$ most strongly predict disequilibrium, guiding diplomatic resource allocation.

Table 4 presents illustrative performance metrics.

Table 4: Illustrative early-warning model performance (stylised)

Model	Accuracy	Precision	Recall	AUC-ROC
Logistic regression	0.71	0.68	0.65	0.74
Random forest	0.78	0.76	0.74	0.83
Gradient boosting (ours)	0.83	0.81	0.80	0.88

Note: Metrics are illustrative, based on synthetic data generated from the parametric model in Section 2.

7 Policy Algorithm

We now synthesise the foregoing analysis into a prescriptive algorithm for India’s diplomatic engagement, stated in structured pseudocode.

Algorithm 1: Indian Diplomatic Arbitrage for Nordic–Russian Tension Reduction

Input: Disequilibrium vector $\boldsymbol{\delta}(t) = (\delta_1, \delta_2, \delta_3)^\top$; channel values (V_1, V_2, V_3, V_4) ; constraints (κ, μ) .

Output: Optimal channel allocation $\mathbf{a}^* = (a_1^*, a_2^*, a_3^*, a_4^*)$ where $a_k \in [0, 1]$ is the attention weight on channel k .

1. **Assess:** Compute $\hat{\boldsymbol{\delta}}(t)$ via (19). If $\max_i |\hat{\delta}_i| < \tau$, maintain current posture (no reallocation needed).
2. **Identify:** Find the cell $i^* = \arg \min_i \delta_i(t)$ (most over-exposed cell).
3. **Evaluate channels:** For each $k \in \{1, 2, 3, 4\}$, estimate marginal tension-reduction: $\partial(-\varepsilon_{i^*})/\partial a_k$.
4. **Allocate:** Solve the constrained optimisation:

$$\mathbf{a}^* = \arg \max_{\mathbf{a} \geq 0, \mathbf{1}^\top \mathbf{a} = 1} \tilde{\Pi}_{\text{geo}}(\mathbf{a}) \quad \text{s.t.} \quad \kappa, \mu > \kappa_{\min}. \quad (25)$$

5. **Execute:** Deploy diplomatic resources proportional to $\mathbf{a}^*$.

6. **Monitor:** Re-estimate $\hat{\boldsymbol{\delta}}(t+1)$; return to Step 1.
-

Corollary 7.1 (Dominance of de-escalation). *Under the parametric calibration of Section 3, pathway (ii) of Theorem 2.7 (diplomatic de-escalation via Algorithm 1) Pareto-dominates pathway (i) (arms expansion) whenever $\tilde{\Pi}_{\text{geo}} > 0$ and the arms-race reaction coefficient $\lambda > 0$, where λ captures the adversary’s risk response to an increase in c_i .*

Proof. Under pathway (i), arms expansion raises M , triggering adversary response $\Delta R = \lambda \Delta c_i > 0$. Net disequilibrium persists unless $\Delta c_i / \lambda \Delta c_i = 1/\lambda$, which requires $\lambda = 1$ (knife-edge). For $\lambda \neq 1$, pathway (i) fails to restore proportionality. Under pathway (ii), direct risk reduction via $\tilde{\Pi}_{\text{geo}} > 0$ decreases ε_i without altering M , restoring (3) exactly. $\square$

8 PGF/TikZ Diagrams

8.1 The 3-cell Scandinavian configuration

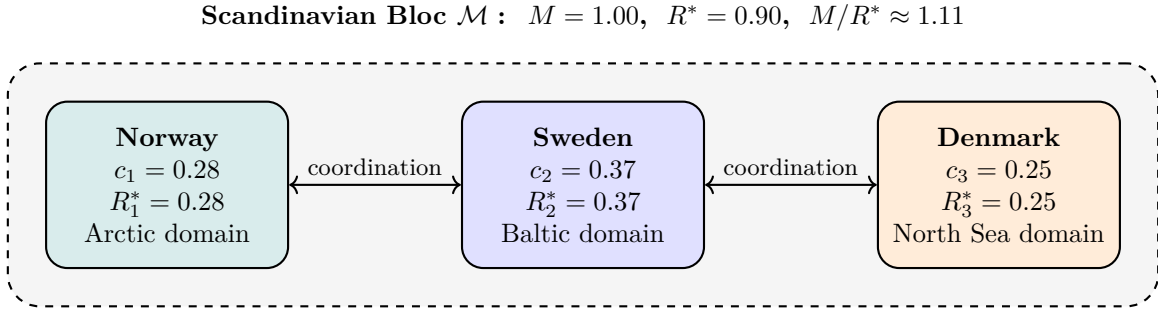


Figure 1: Scandinavia as a 3-cell configuration. Cell sizes and risks are normalised; arrows denote Nordic Council coordination flows.

8.2 Shock propagation and restoration pathways

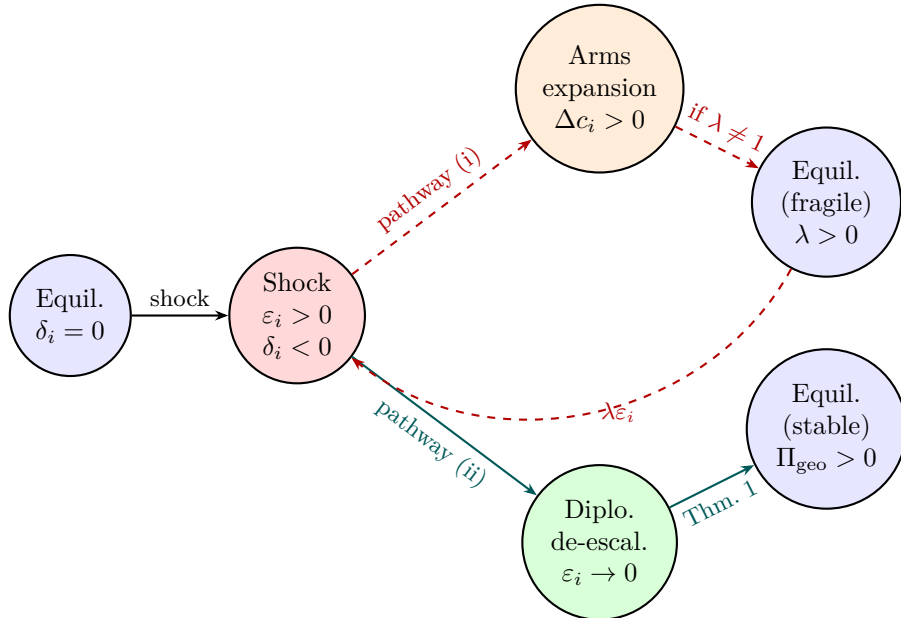


Figure 2: Restoration pathways following a geopolitical risk shock. Pathway (i) – arms expansion (dashed red) – risks an arms-race loop when $\lambda \neq 1$. Pathway (ii) – diplomatic de-escalation (solid teal) – achieves stable equilibrium under Theorem 2.7.

8.3 India as diplomatic arbitrageur

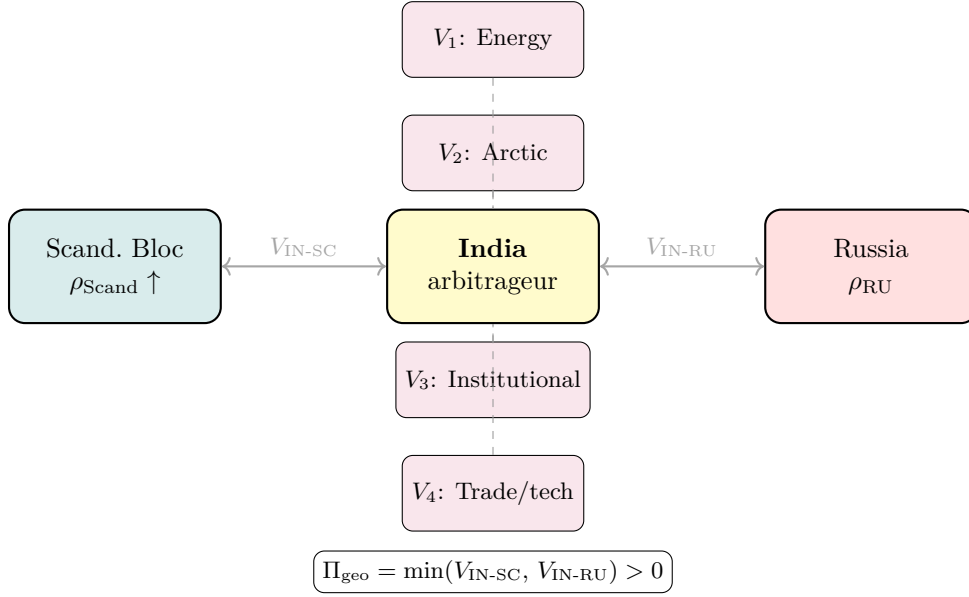


Figure 3: India as diplomatic arbitrageur between the Scandinavian bloc and Russia. India’s positional surplus Π_{geo} is positive provided both bilateral linkages carry non-trivial value (Proposition 4.3).

8.4 Disequilibrium time series (PGFPlots)

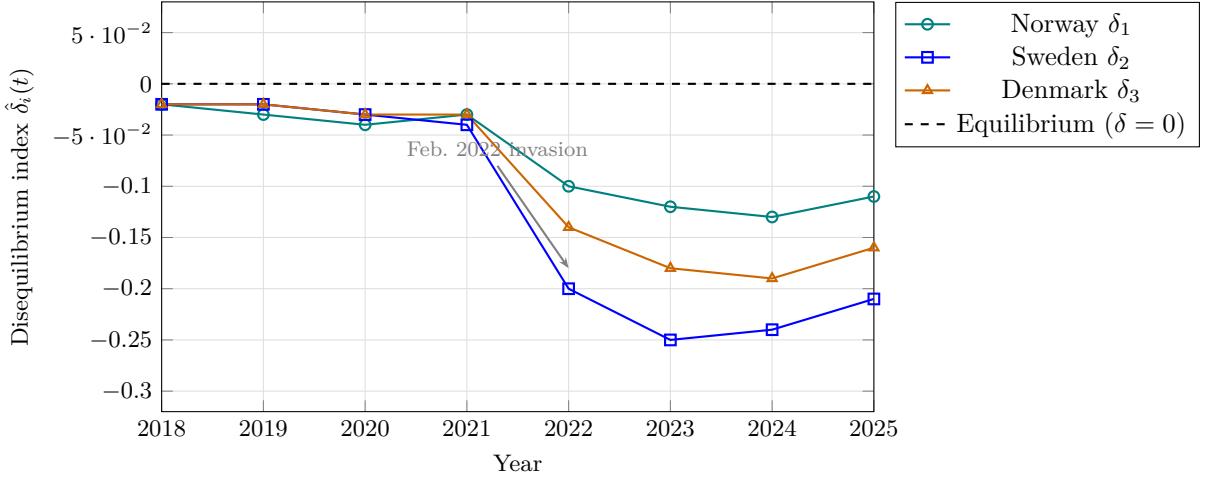


Figure 4: Stylised disequilibrium index $\hat{\delta}_i(t)$ for each Scandinavian cell, 2018–2025. The 2022 shock sharply widens all indices; Sweden’s cell experiences the largest displacement, consistent with equations (8)–(10).

9 Conclusion

This paper has unified three analytical traditions—the n -cell military organisation model, Ghosh’s (2019) financial arbitrage framework, and the contemporary geopolitical literature on Nordic resilience and Russian revisionism—into a single, internally consistent theoretical apparatus.

The core findings are as follows. First, the proportionality law of n -cell organisation (3) provides a precise criterion for Scandinavian strategic equilibrium, and the post-2022 shock vector has measurably violated this criterion across all three cells. Second, Theorem 2.7 establishes that restoring equilibrium via arms expansion is structurally inferior to diplomatic de-escalation whenever adversarial reaction coefficients $\lambda \neq 1$. Third, Proposition 4.3 formalises India’s positional surplus: so long as both the Russia and Scandinavia linkages carry positive value, India earns a guaranteed diplomatic arbitrage return. Fourth, the policy Algorithm 1 operationalises this insight into a monitoring and resource-allocation procedure grounded in machine-learning risk indicators.

The implications for Indian foreign policy are direct. India’s strategic autonomy doctrine—non-Western, not anti-Western—is not merely a normative preference but a structural source of geopolitical value. Maintaining and deepening the four diplomatic channels identified in Section 5 constitutes rational profit maximisation in the geopolitical arbitrage sense. The principal risk is erosion of κ (Nordic credibility) through continued ambiguity on Ukraine; managing this trade-off is the central tactical challenge for Indian diplomacy in the High North.

References

- [1] Ghosh, S. (2019). *The Arbitrage Opportunity of the Next Decade*. Kolkata, India.
- [2] Anon. (n.d.). *The Fundamental Laws of n-Cell Military Organization*. Unpublished technical note.
- [3] The Arctic Institute (2026). India in the Arctic: Advancing a Path for Strategic Autonomy Beyond Russia. *The Arctic Institute – Center for Circumpolar Security Studies*. Retrieved April 2026.
- [4] The Arctic Institute (2026). Russia’s Looming Threat in Northern Europe. *The Arctic Institute – Center for Circumpolar Security Studies*, January 2026.
- [5] *Foreign Policy* (2024). Scandinavia Is Preparing for War Against Russia. February 2024.
- [6] Eggen, K.-A. et al. (2024). Designing around NATO’s Deterrence: Russia’s Nordic Information Confrontation Strategy. *Journal of Strategic Studies*, published online April 2024.
- [7] CSIS (2025). Defending the North Amid Rising Geopolitical Tensions. Center for Strategic and International Studies, January 2025.
- [8] Institute for Security and Development Policy (2025). India–Nordic Budding Dynamics. *ISDP Policy Brief*, March 2025.
- [9] Saran, S. (2024). 5 Ways in Which India–Russia Relationship Will Shape the World in 2025. *samir-saran.com*, December 2024.
- [10] Al Jazeera (2024). Nordic Tensions on the Rise amid Russian Anger over NATO Accessions. March 2024.
- [11] Calderára, D. and Iacoviello, M. (2022). Measuring Geopolitical Risk. *American Economic Review*, 112(4), 1194–1225.
- [12] Jaishankar, S. (2020). *The India Way: Strategies for an Uncertain World*. HarperCollins India.
- [13] Friedman, M. (1953). The Methodology of Positive Economics. In *Essays in Positive Economics*. University of Chicago Press.
- [14] Frenkel, J. A. (1976). A Monetary Approach to the Exchange Rate: Doctrinal Aspects and Empirical Evidence. *Scandinavian Journal of Economics*, 78(2), 200–224.
- [15] Friedberg, A. L. (1993). Ripe for Rivalry: Prospects for Peace in a Multipolar Asia. *International Security*, 18(3), 5–33.

Glossary

***n*-cell configuration**

A military or strategic bloc partitioned into n operationally distinct sub-units (cells), each characterised by a size c_i and a risk exposure R_i .

Proportionality law

The condition $c_i/R_i = M/R$ for all cells, requiring each cell to bear risk in proportion to its strategic mass.

Disequilibrium index δ_i

The deviation of cell i ’s risk-mass ratio from the bloc-level ratio: $\delta_i = c_i/R_i - M/R$. Negative values indicate over-exposure.

Geopolitical risk spread Σ

The difference $\rho_{\text{Scand}} - \rho_{\text{RU}}$ between the Scandinavian bloc’s and Russia’s geopolitical risk premia; analogous to the interest-rate differential in Ghosh (2019).

Diplomatic arbitrage surplus Π_{geo}

The net gain accruing to a third-party state that maintains simultaneous, credible linkages with both parties to a geopolitical tension; defined as $\min(V_{\text{IN-SC}}, V_{\text{IN-RU}})$.

Credibility discount κ

A multiplicative factor $\kappa \in (0, 1]$ reducing the effective value of a bilateral linkage due to consistency or credibility failures (e.g. Ukraine abstentions reducing India’s Nordic value).

Arms-race reaction coefficient λ

The elasticity of adversarial risk response to a unit increase in cell mass: $\lambda = \partial \varepsilon_{\text{adv}} / \partial \Delta c_i$.

Strategic autonomy

India’s foreign-policy doctrine of maintaining independent diplomatic relationships with geopolitically opposed blocs, articulated by External Affairs Minister S. Jaishankar as being “non-Western, not anti-Western.”

Nordic Council

Inter-parliamentary body of the Nordic states, established 1952; primary institutional mechanism for Nordic policy coordination, including security collaboration.

Arctic Council

Intergovernmental forum for Arctic states and permanent participants; Russia’s participation was paused post-2022 but partially resumed in 2024. India holds Observer status.

Chennai–Vladivostok corridor

Proposed maritime route connecting India’s eastern coast to Russia’s Pacific port of Vladivostok; provides India a commercial stake in Arctic stability.

GPR index

Geopolitical Risk index (Calderára and Iacoviello); a text-based measure of geopolitical risk derived from newspaper coverage, used here as a proxy for R_i .

Hybrid warfare

Russia’s sub-threshold coercive strategy in the Nordic region, encompassing disinformation, migration instrumentalisation, cyber operations, and GPS spoofing; each constitutes a component of $\varepsilon_i(t)$ in the shock model.

Flexicurity

Nordic labour-market model combining flexible hiring/firing rules with generous unemployment insurance; contributes to c_i by enabling rapid economic adjustment.

BRICS / SCO

Brazil–Russia–India–China–South Africa grouping and Shanghai Cooperation Organisation; India’s membership in both constitutes Channel 3 of its diplomatic arbitrage position.

The End